

Multimedia Computing

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- ▶ Blurring
- ▶ Sharpening
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- ▶ First order derivative
- ▶ Compass gradient operators
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- ▶ Color edge detection
- ▶ Median filtering
- ▶ Minimum/Maximum filtering

Image mask

- ▶ A mask is a filter. Concept of masking is also known as spatial filtering. we just deal with the filtering operation that is performed directly on the image.

Filtering

- ▶ The process of filtering is also known as convolving a mask with an image. As this process is same of convolution so filter masks are also known as convolution masks.

Types of filters

- linear filters or smoothing filters
- others are called as frequency domain filters.

Filtering

- Filters are used for Blurring and noise reduction
- Filters are used or edge detection and sharpness

Edges

- ▶ Edges are significant local changes of intensity in a digital image. An edge can be defined as a set of connected pixels that forms a boundary between two disjoint regions. There are three types of edges:
- ▶ Horizontal edges
- ▶ Vertical edges
- ▶ Diagonal edges

Edge detection

- ▶ Edge Detection is a method of segmenting an image into regions of discontinuity. It is a widely used technique in digital image processing like
 - ▶ pattern recognition
 - ▶ image morphology
 - ▶ feature extraction

Edge detection

- ▶ Edge detection allows users to observe the features of an image for a significant change in the gray level. This texture indicating the end of one region in the image and the beginning of another. It reduces the amount of data in an image and preserves the structural properties of an image.
- ▶ Edge Detection Operators are of two types:
- ▶ **Gradient** - based operator which computes first-order derivations in a digital image like, Sobel operator, Prewitt operator, Robert operator
- ▶ **Gaussian** - based operator which computes second-order derivations in a digital image like, Canny edge detector, Laplacian of Gaussian

Histogram statistics for image enhancement

Use of histogram statistics for image enhancement

- r denotes a discrete random variable
- $p(r_i)$ denotes the normalized histogram component corresponding to the i th value of r

- Mean
$$m = \sum_{i=0}^{L-1} r_i p(r_i)$$

Histogram statistics for image enhancement

- The nth moment
$$\mu_n(r) = \sum_{i=0}^{L-1} (r_i - m)^n p(r_i)$$

- The second moment
$$\mu_2(r) = \sum_{i=0}^{L-1} (r_i - m)^2 p(r_i)$$

- ▶ Global enhancement: The global mean and variance are measured over an entire image
- ▶ Local enhancement: The local mean and variance are used as the basis for making changes

Convolution

► Convolution (Mathematical formulation)

$$g(x, y) = h(x, y) * f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x - x', y - y') f(x', y') dx' dy'$$

$$y(m, n) = h(m, n) * x(m, n) = \sum_{m'=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} h(m - m', n - n') x(m', n')$$

► Discrete Convolution

- A convolution is merely a weighted sum of pixels in the neighborhood of the source pixel.
- The weights are determined by a small matrix called the **convolution mask** or **convolution kernel**.
- The new pixel value is computed by multiplying each pixel value in the neighborhood with the corresponding weight in the convolution mask and summing these products.
- The location of the center corresponds to the location of the output pixel
- Image smoothing, edge detection, and other effects.

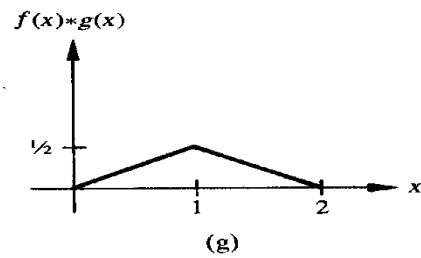
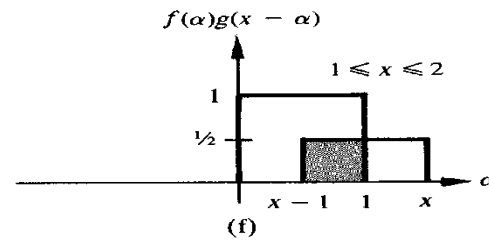
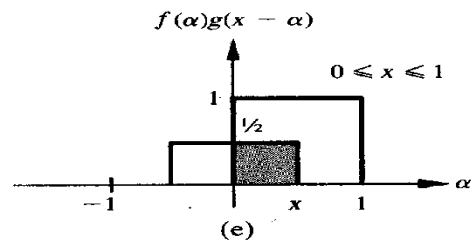
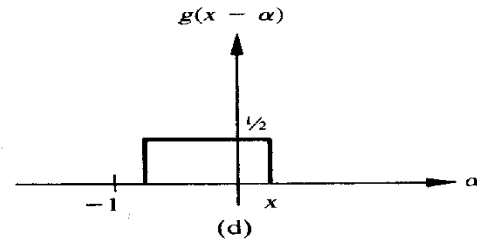
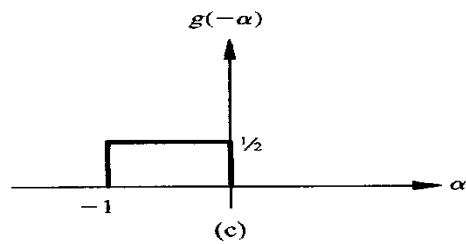
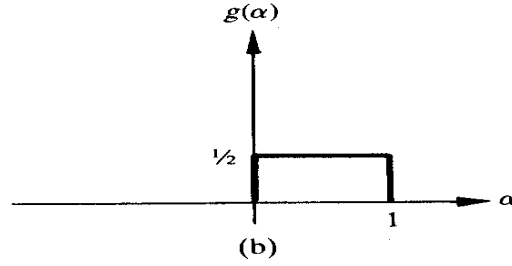
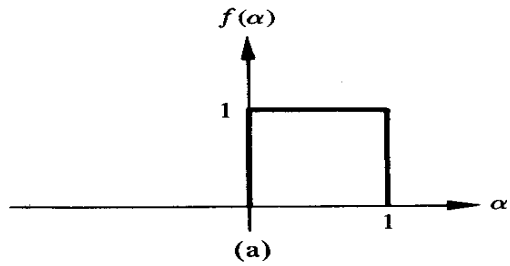
1D Convolution

■ 1D Convolution

- Convolution is very important to understand image processing
- A basic operation of linear systems theory
- Convolution of two function $f(x)$ and $g(x)$

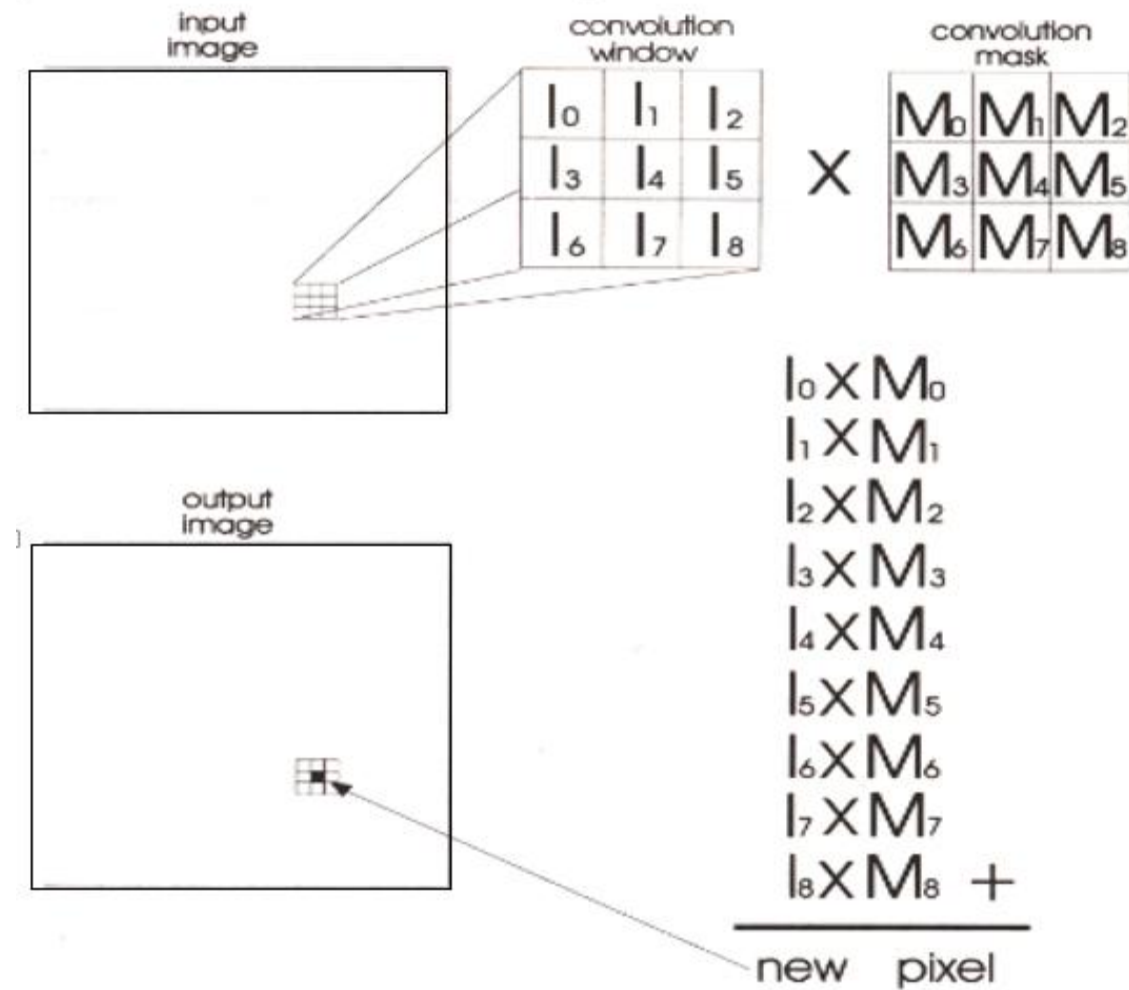
$$f(x) * g(x) = \int_{-\infty}^{\infty} f(\alpha)g(x-\alpha)d\alpha$$

- Intuitively, one function is swept past (1D) or rubbed over (2D) the other.



$$f(x) * g(x) = \begin{cases} x/2 & 0 \leq x \leq 1 \\ 1 - x/2 & 1 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

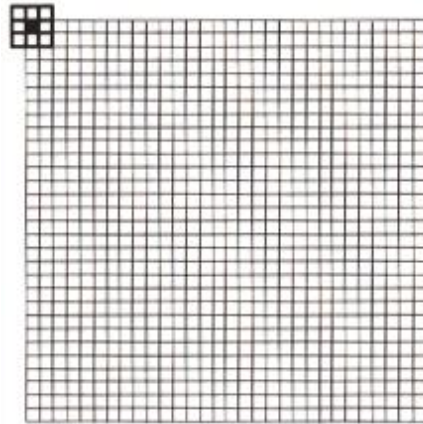
Convolution



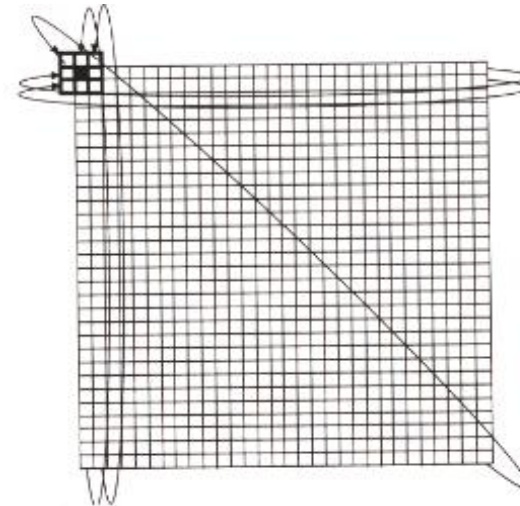
Convolution

□ How to treat image borders for convolution ?

- **Zero-padding:** treat the empty cells in the convolution window as zeros.
- **Start convolving at the first position where window doesn't overlap the image.**
 - Ex. Start convolving at (1,1) instead of (0,0) with 3 x 3 convolution mask.
- **Duplication of the edges**
 - Ex. Copy the top row, the left side, the right side, and the bottom of the image.
- **Wrap-around**



Duplication of the
edges



Wrap-around

Blurring

► Blurring

► Blurring (low-pass spatial filtering)

- Removes the finer details of an image.
- Can be accomplished through convolution.

► One common blurring mask

- all convolution coefficients are equal.
- In a 3 X 3 mask, all the elements equal $1/9$,
- In a 5 X 5 mask, all elements equal $1/25$.
- The bigger the mask, the greater the blurring effect.

3 X 3 MASK

```
1/9 1/9 1/9
1/9 1/9 1/9
1/9 1/9 1/9
```

5 X 5 MASK

```
1/25 1/25 1/25 1/25 1/25
1/25 1/25 1/25 1/25 1/25
1/25 1/25 1/25 1/25 1/25
1/25 1/25 1/25 1/25 1/25
1/25 1/25 1/25 1/25 1/25
```


Blurring



Original Image



3X3 blurring

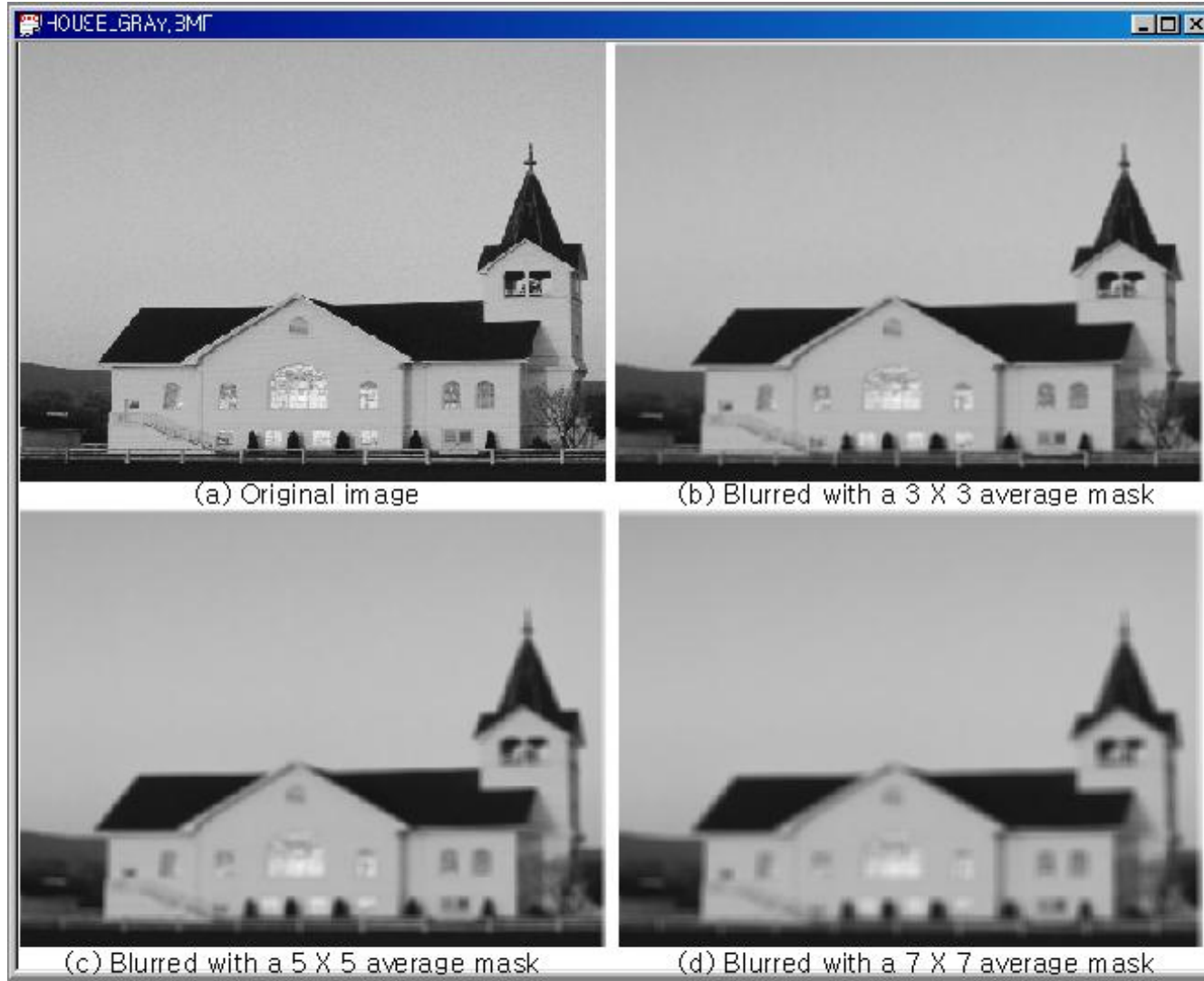


5X5 blurring



7X7 blurring

Blurring Example

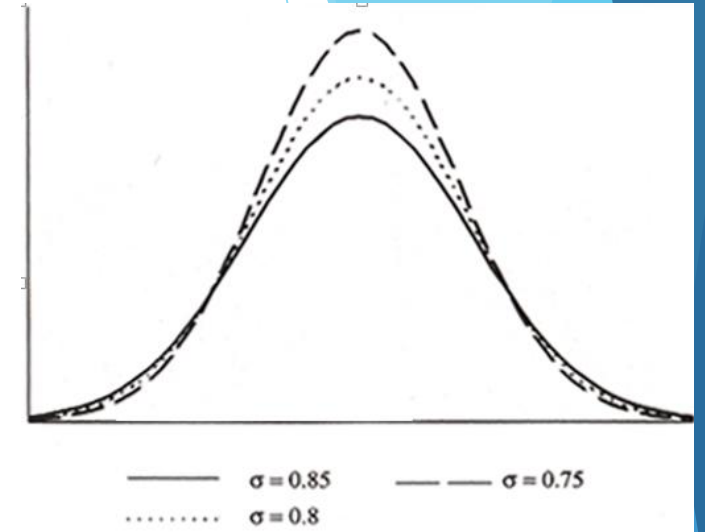


Gaussian Smoothing Filters for Burring

- ▶ Gaussian smoothing filters & mask approximation

$$G[x,y] = \frac{e^{\frac{-(x^2+y^2)}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}$$
$$\begin{bmatrix} 1/16 & 1/8 & 1/16 \\ 1/8 & 1/4 & 1/8 \\ 1/16 & 1/8 & 1/16 \end{bmatrix}$$

σ controls the effective spread & size of the mask.



Original Image



Gaussian Filtering

Sharpening

▶ Sharpening

- ▶ Sharpening has the opposite affect of blurring.
- ▶ Sharpening, also referred to as crispening, emphasizes the details in an image.
- ▶ Sharpening an image will increase its contrast.
- ▶ Some sharpening mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix} \quad \begin{bmatrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & -2 & 1 \\ -1 & 5 & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

Sharpening(Cont'd)

▶ High pass Filter

- ▶ Sharpening an image with the high filter.

- ▶ A high pass filter will remove the low frequency components and show only high detail.
- ▶ A common high pass filter mask:

$$\begin{bmatrix} -1/9 & -1/9 & -1/9 \\ -1/9 & 8/9 & -1/9 \\ -1/9 & -1/9 & -1/9 \end{bmatrix}$$

▶ Unsharp masking : another method of highpass filtering

- ▶ Subtract a lowpass filtered image from its original.

▶ High-boost filtering:

High - Boost = α Original - Lowpass

When $\alpha = 1$, the result is highpass filter.

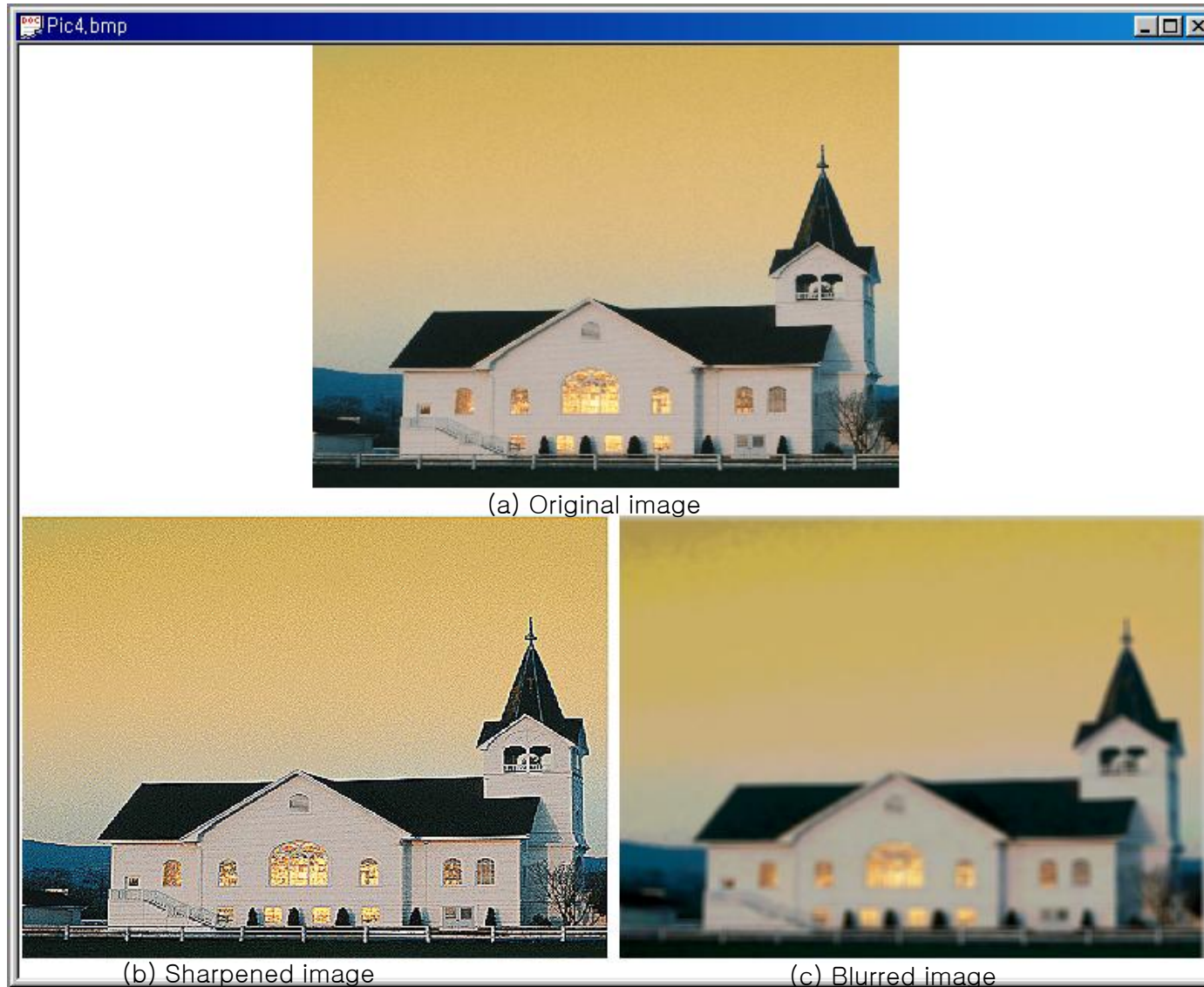
When $\alpha > 1$, the effect is to restore low frequency Component.

$$\begin{bmatrix} -1/9 & -1/9 & -1/9 \\ -1/9 & \omega/9 & -1/9 \\ -1/9 & -1/9 & -1/9 \end{bmatrix}, \omega = 9\alpha - 1$$

Sharpening Example



Color Sharpening & Blurring Example



High-boost filtering

- ▶ High-boost filtering(high-frequency emphasis filter):

High - Boost = α Original - Lowpass

When $\alpha = 1$, the result is high pass filter.

When $\alpha > 1$, the effect is to restore low frequency component

$$\begin{bmatrix} -1/9 & -1/9 & -1/9 \\ -1/9 & \omega/9 & -1/9 \\ -1/9 & -1/9 & -1/9 \end{bmatrix}, \omega = 9\alpha - 1$$

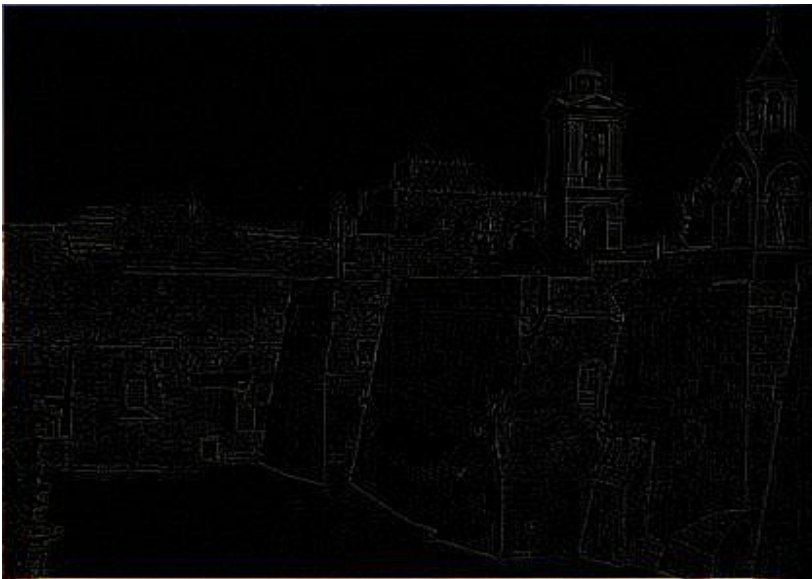
- ▶ The effect of high-boost filtering:
 - ▶ Retains more of original image's background
 - ▶ As α increases, the overall image gets lighter.
 - ▶ Looks more like the original image, with a relative degree of edge enhancement.



Original Image



Sharpening

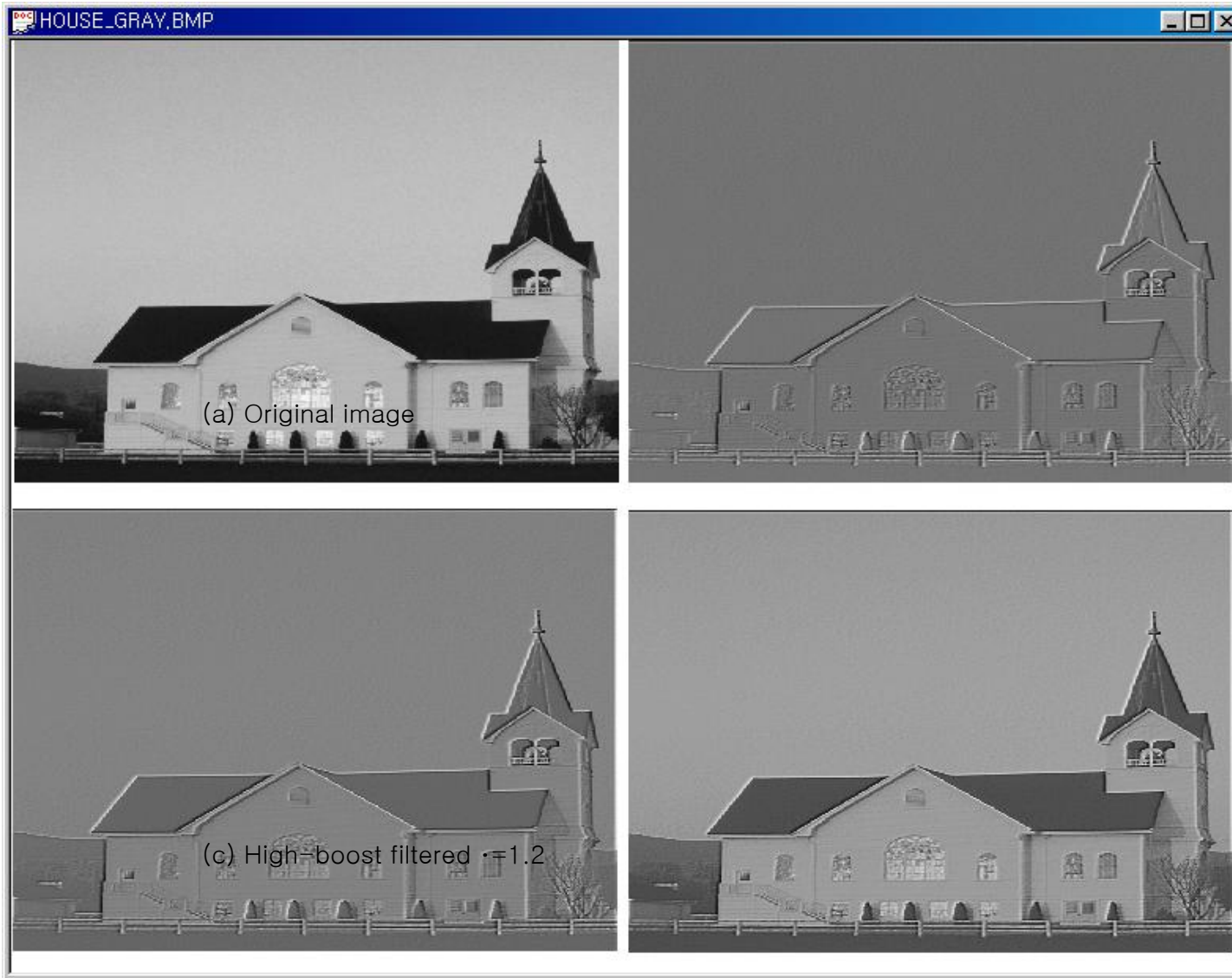


Highpass



• =1.2 High-boost

High pass filter & High-boost filter Example



Detection of discontinuities

- ▶ Three types of discontinuities: points, lines, edges
- ▶ Point detection:
 - ▶ A point has been detected at the location at which the mask is centered if $|R| > T$
 - ▶ Where T is a nonnegative threshold
 - ▶ $R = \sum_{i=1}^9 Z_i M_i$

Z_1	Z_2	Z_3
Z_4	Z_5	Z_6
Z_7	Z_8	Z_9

M_1	M_2	M_3
M_4	M_5	M_6
M_7	M_8	M_9

Detection of discontinuities

- ▶ A mask used for detecting isolated points different from a constant background

-1	-1	-1
-1	8	-1
-1	-1	-1

- ▶ Line masks for line detection

-1	-1	-1
2	2	2
-1	-1	-1

Horizontal

-1	-1	2
-1	2	-1
2	-1	-1

+ 45°

-1	2	-1
-1	2	-1
-1	2	-1

Vertical

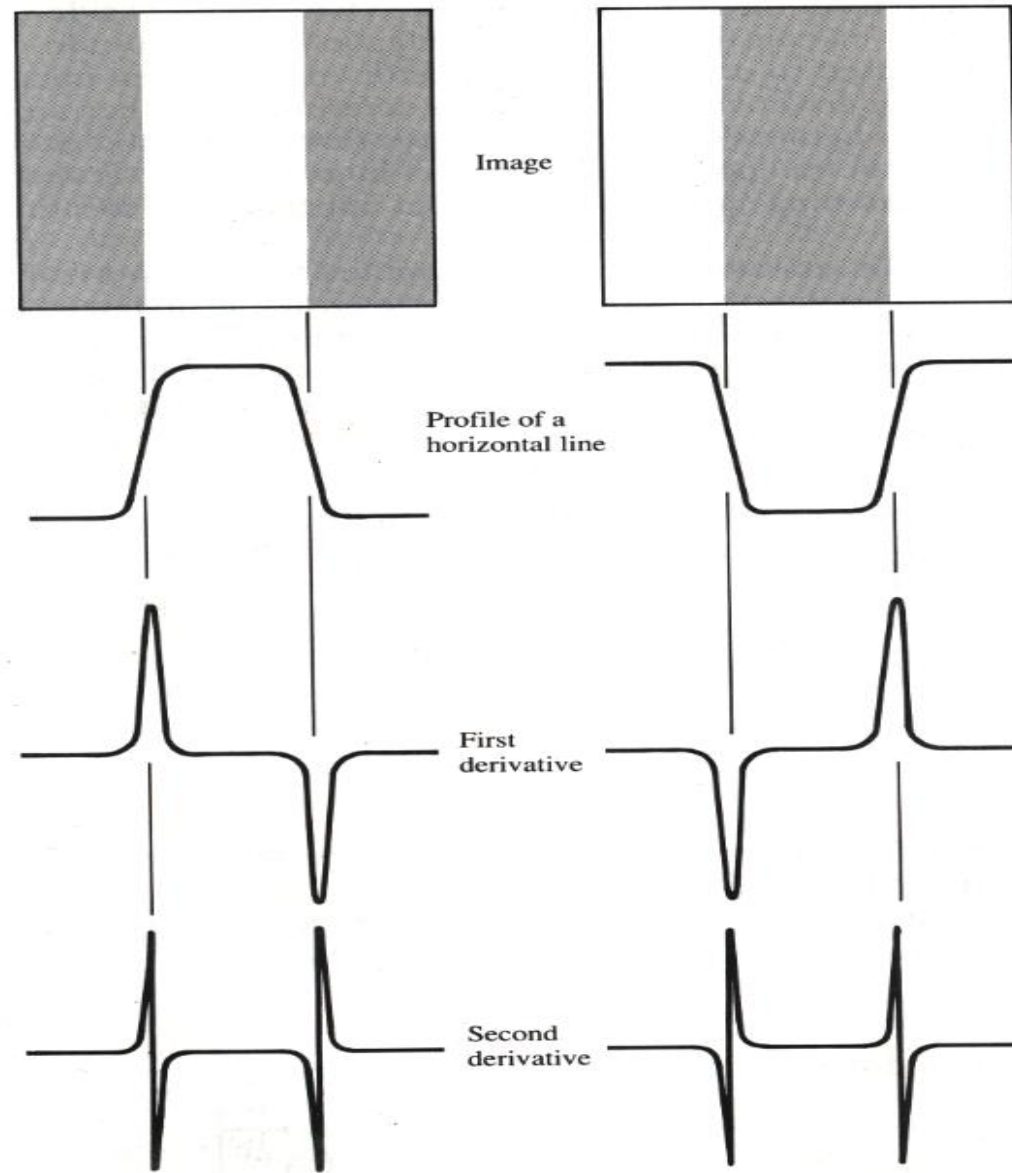
2	-1	-1
-1	2	-1
-1	-1	2

-45°

Edge Detection

- ▶ Isolated points and thin lines are not frequent occurrences in most practical applications.
- ▶ The most common approach for detecting meaningful discontinuities in gray level.
- ▶ Edge: the boundary between two regions with relatively distinct gray-level properties.
 - ▶ Assume that the regions in question are sufficiently homogeneous s. t. the transition between two regions are determined on the basis of gray-level discontinuities alone.
- ▶ The idea underlying most edge-detection techniques
 - ▶ The computation of a local derivative.

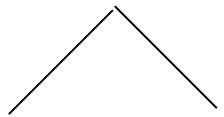
Edge Detection



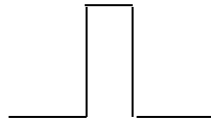
Edge Detection

▶ Edge Detection

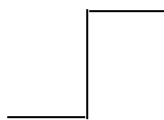
- ▶ The edges of an image: hold much of the information in that image.
 - ▶ The edges tell where objects are, their shape and size, and something about their texture.
 - ▶ An edge is where the intensity of an image move from a low value to a high value or vice versa.



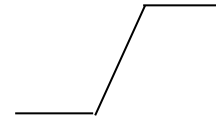
roof edge



line edge



step edge



ramp edge

- ▶ Infinite number of edge orientations, widths and shapes.

- ▶ The edge detection can be used in many applications.
 - ▶ Image segmentation : Image segmentation: a field of image analysis, is used to group pixels into regions to determine an image's composition
 - ▶ Image registration : aligns two images that may have been acquired at separate times or from different sensors.

Edge Detection

The edges of an image hold much of the information in that image

The edges tell

- where objects are
- their shape and size
- something about their texture

Edge Detection

▶ Edge detection techniques

▶ Homogeneity operator

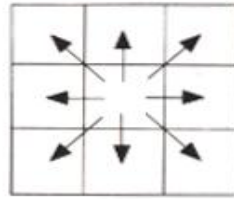
- ▶ The simplest and quickest edge detectors determine the maximum value from a series of pixel subtractions.
- ▶ Subtracts each 8 surrounding pixels from the center pixel of a 3 X 3 window.
- ▶ The output is the maximum of the absolute value of each difference.

▶ Difference operator

- Similar to the homogeneity operator is the difference edge detector.
- It operates more quickly because it requires four subtractions per pixel as opposed to the eight needed by the homogeneity operator.

Edge Detection

homogeneity operator

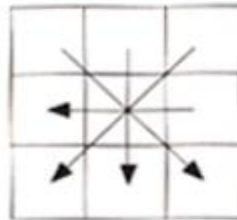


image

11	13	15
16	11	11
16	12	11

$$\text{new pixel} = \text{maximum of } \{ |11-11| \quad |11-13| \quad |11-15| \\ |11-16| \quad |11-11| \\ |11-16| \quad |11-12| \quad |11-11| \} = 5$$

difference operator



image

11	13	15
16	11	11
16	12	11

$$\text{new pixel} = \text{maximum of } \{ |11-11| \quad |13-12| \quad |15-16| \quad |11-16| \} = 5$$

Edge Detection

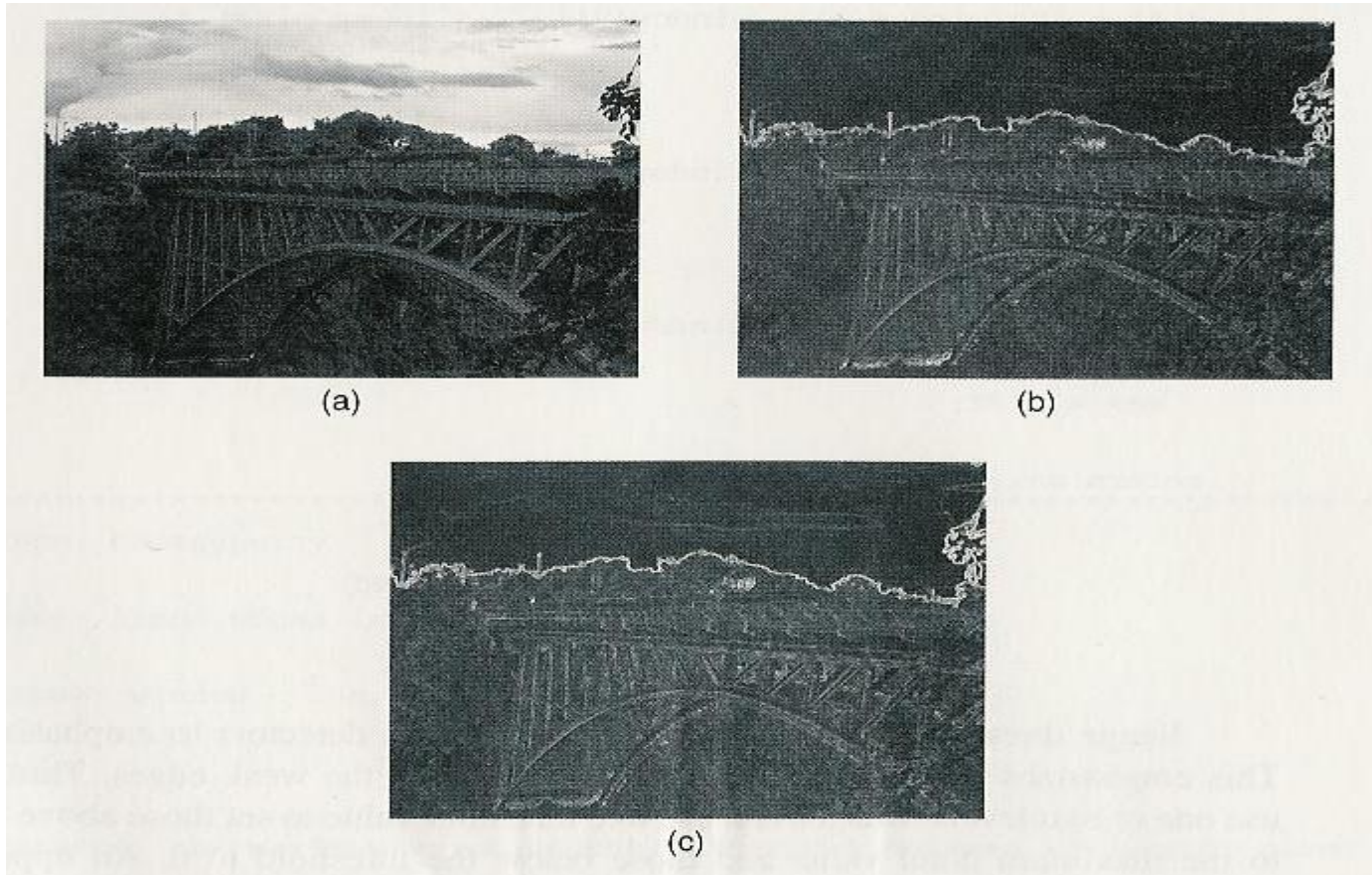
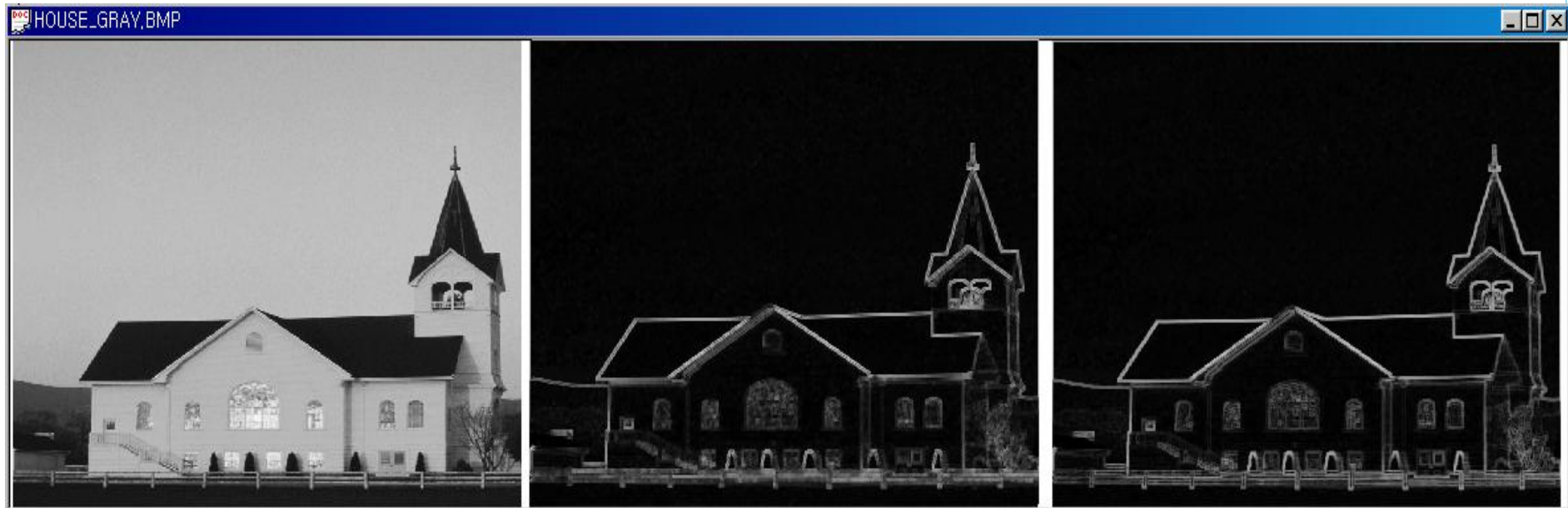


Fig. 3.14 (a) Original image, (b) edge map of homogeneity operator, and (c) difference operator.

Edge Detection



(a) Original image

(b) Edge map of homogeneity operator

(c) Difference operator

Derivative Operator

► First Derivative Operator

$$H(x, y) = \sqrt{H_r^2(x, y) + H_c^2(x, y)}$$

or

$$H(x, y) = |H_r(x, y)| + |H_c(x, y)|$$

$$\theta = \tan^{-1} \frac{H_c(x, y)}{H_r(x, y)}$$

	H_r	H_c
Roberts Operator	$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Prewitt Operator	$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$	$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$
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	H_r	H_c
Sobel Operator	$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$	$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

Frei-Chen Operator	$\begin{bmatrix} 1 & 0 & -1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & 0 & -1 \end{bmatrix}$	$\begin{bmatrix} -1 & -\sqrt{2} & -1 \\ 0 & 0 & 0 \\ 1 & \sqrt{2} & 1 \end{bmatrix}$
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Derivative Operator

- ▶ Roberts operator
 - ▶ Smaller effective area than the other masks
 - ▶ More susceptible to noise
- ▶ Sobel operator
 - ▶ More sensitive to diagonal edges than vertical and horizontal edges.
- ▶ Prewitt operator
 - ▶ More sensitive to vertical and horizontal edges than diagonal edges

Derivative Operator

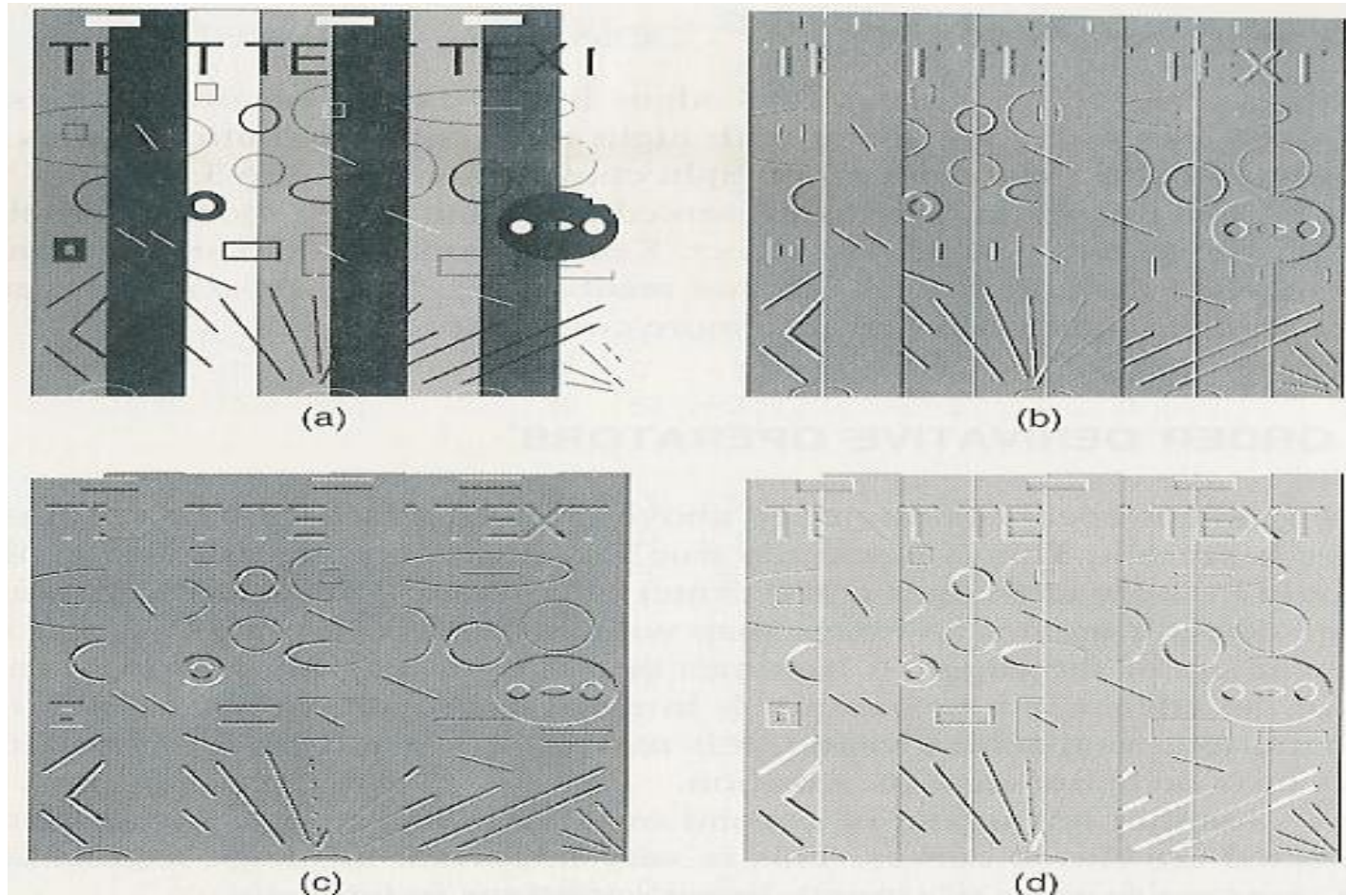


Fig. 3.17 (a) Original image, (b) Hr convolved, (c) Hc convolved, and (d) composite.

Derivative Operator

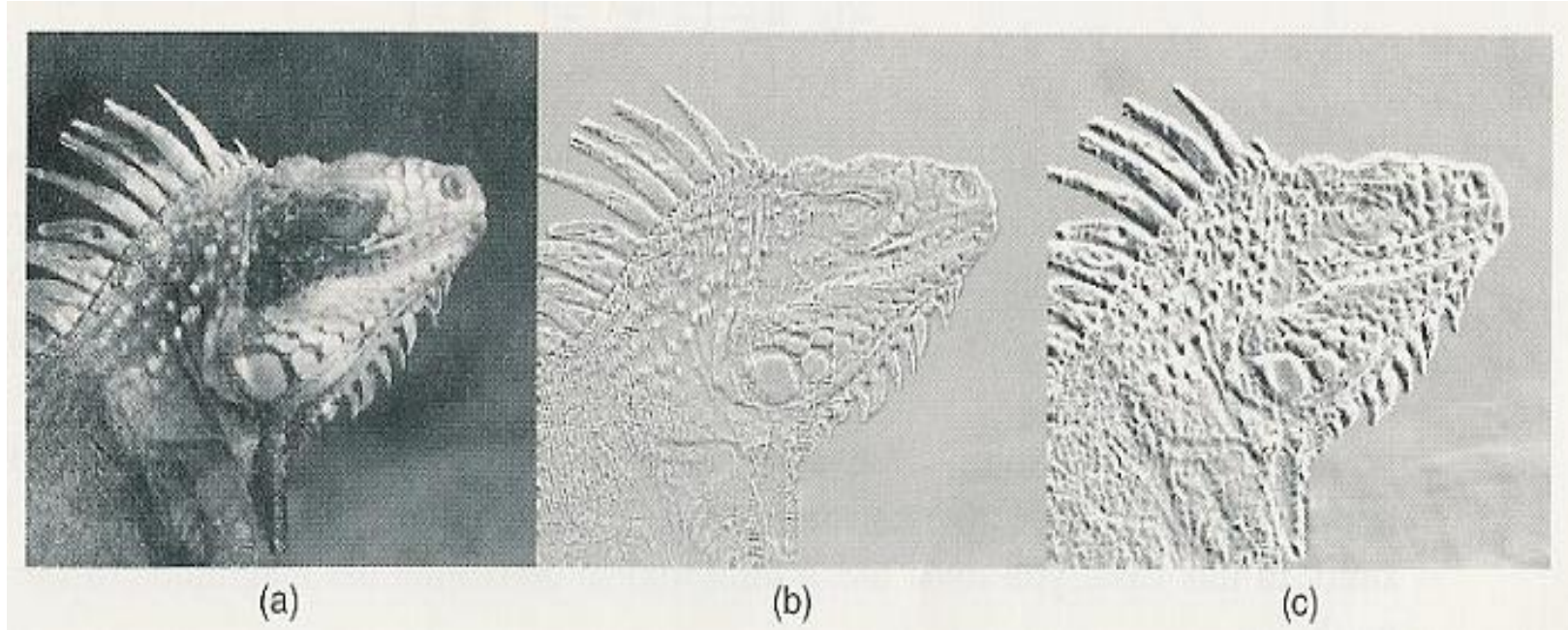


Fig. 3.18 (a) Original image, (b) processed with the Roberts operator, and (c) processed with the Prewitt operator.

Compass Gradient Operator

- ▶ Compass Gradient Operator
 - ▶ Find edges in eight different orientations.
 - ▶ Convolve an image with eight different masks.
 - ▶ Prewitt, Kirsh, Robinson 3-level, Robinson 5-level
- ▶ The effects of convolution
 - ▶ Smaller masks are sensitive to noise
 - ▶ Larger masks cannot resolve fine detail, and also
 - ▶ Requires more computations.

Compass Gradient Operator

	Prewitt	Kirsch	Robinson 3-level	Robinson 5-level
	$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & -1 \\ 1 & 1 & -1 \end{bmatrix}$	$\begin{bmatrix} 5 & -3 & -3 \\ 5 & 0 & -3 \\ 5 & -3 & -3 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$
$\begin{bmatrix} -1 & -1 \\ -1 & -2 \\ 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & -1 & -1 \\ 1 & -2 & -1 \\ 1 & 1 & -1 \end{bmatrix}$	$\begin{bmatrix} -3 & -3 & -3 \\ 5 & 0 & -3 \\ 5 & 5 & -3 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}$
$\begin{bmatrix} -1 & 1 \\ -1 & -2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & -1 & -1 \\ 1 & -2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} -3 & -3 & -3 \\ -3 & 0 & -3 \\ 5 & 5 & 5 \end{bmatrix}$	$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & -1 \end{bmatrix}$
	$\begin{bmatrix} -3 & 0 & 5 \\ -3 & -3 & 5 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$	
$\begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & 1 \\ -1 & -1 & 1 \end{bmatrix}$	$\begin{bmatrix} -3 & 5 & 5 \\ -3 & 0 & 5 \\ -3 & -3 & -3 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \\ -2 & -1 & 0 \end{bmatrix}$	
$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & 1 \\ -1 & -1 & -1 \end{bmatrix}$	$\begin{bmatrix} 5 & 5 & 5 \\ -3 & 0 & -3 \\ -3 & -3 & -3 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$	
$\begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & 1 \\ -1 & -1 & 1 \end{bmatrix}$	$\begin{bmatrix} 5 & 5 & -3 \\ 5 & 0 & -3 \\ -3 & -3 & -3 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & -1 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & -2 \end{bmatrix}$	

Sobel Operator

- Sobel Operator: It is a discrete differentiation operator. It computes the gradient approximation of image intensity function for image edge detection. At the pixels of an image, the Sobel operator produces either the normal to a vector or the corresponding gradient vector. It uses two 3 x 3 kernels or masks which are convolved with the input image to calculate the vertical and horizontal derivative approximations respectively

$$M_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad M_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Sobel operator

Advantages:

- ▶ Simple and time efficient computation
- ▶ Very easy at searching for smooth edges

Limitations:

- ▶ Diagonal direction points are not preserved always
- ▶ Highly sensitive to noise
- ▶ Not very accurate in edge detection
- ▶ Detect with thick and rough edges does not give appropriate results

Second Order Derivative Operator

- ▶ Second Derivative Operator:

- ▶ Better edge localization
- ▶ Laplacian Operator
- ▶ The resulting image exhibits a sign change at image edges => Zero crossing

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \text{ or } \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

- ▶ Disadvantage of Laplacian

- ▶ Typically unacceptably sensitive to noise.
- ▶ Produce double edges
- ▶ Unable to detect edge direction

Zero Crossing Detection

- ▶ Zero Crossing offers an alternative in case when edges are blurry or a high noise content is present.
- ▶ At ZC, generate the output pixels for the edge map.
- ▶ Zero Crossing Detection

```
for(i=0; i<(cols * (rows-1)); i++)  
if((image[i] < 0) && ((image[i+1] >= 0 || (image[i+cols] >=0)))  
    out_buffer[i] = 255; /* Must check both Horizontal and vertical  
else if((image[i] > 0) && ((image[i+1] <= 0 || (image[i+cols] <=0)))  
    out_buffer[i] = 255;  
else  
    out_buffer[i] = 0;
```

Zero Crossing Detection

- ▶ Simple zero-crossing
 - ▶ may indicate more edges than desired.
 - ▶ A method of determining ZC with threshold:
 - ▶ Must use threshold to determine the magnitude of the edge to mask .
 - ▶ Pass a 3x3 window; determine the difference between the maximum and minimum; An edge is present if the difference exceeds the threshold.

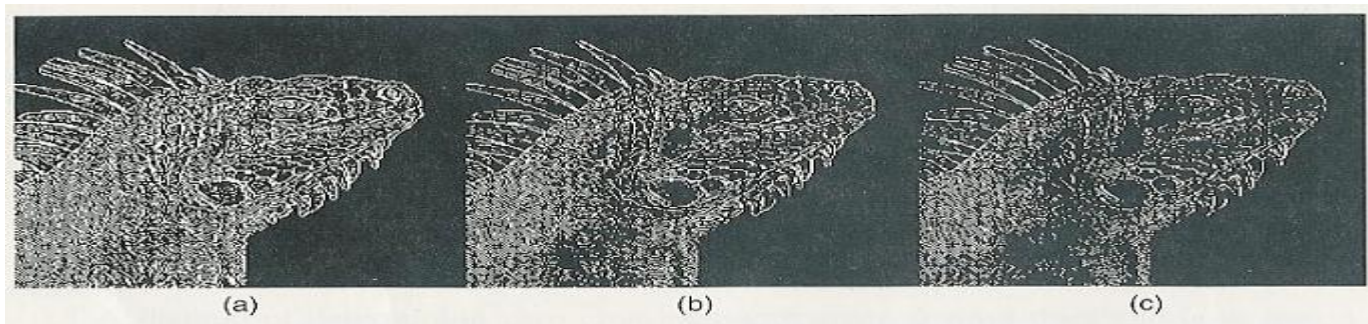
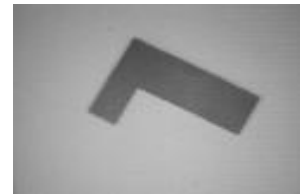
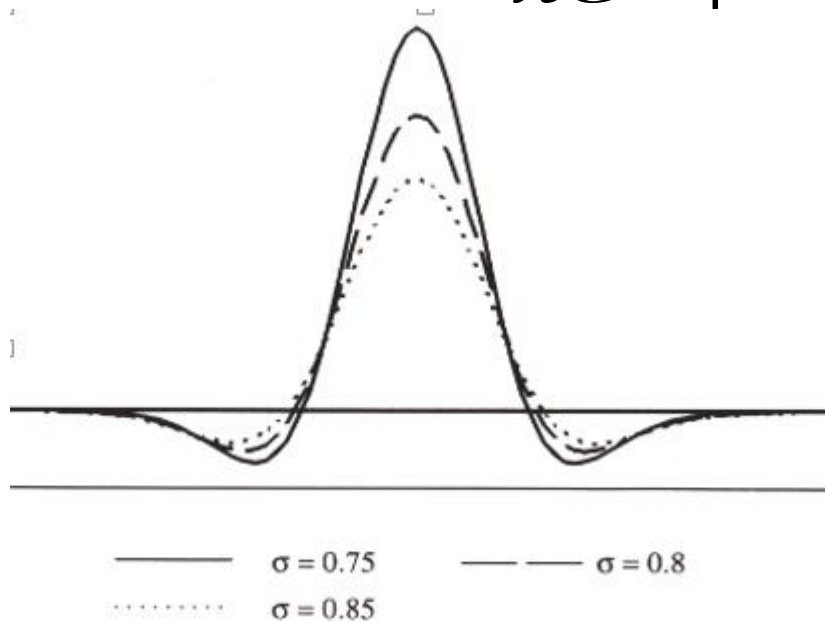


Fig. 3.20 Different thresholds (a) 20, (b) 48, and (c) 72.

LoG(Laplacian of Gaussian)

- ▶ LoG(Laplacian of Gaussian)
 - ▶ A second order derivative edge detector less susceptible to noise
 - ▶ Gaussian smoothing before applying Laplacian

$$LoG(x, y) = \frac{1}{\pi\sigma^4} \left[1 - \frac{(x^2 + y^2)}{2\sigma^2} \right] - e^{\frac{-(x^2 + y^2)}{2\sigma^2}}$$



Input Image



7X7, $\sigma=1.0$



ut Image

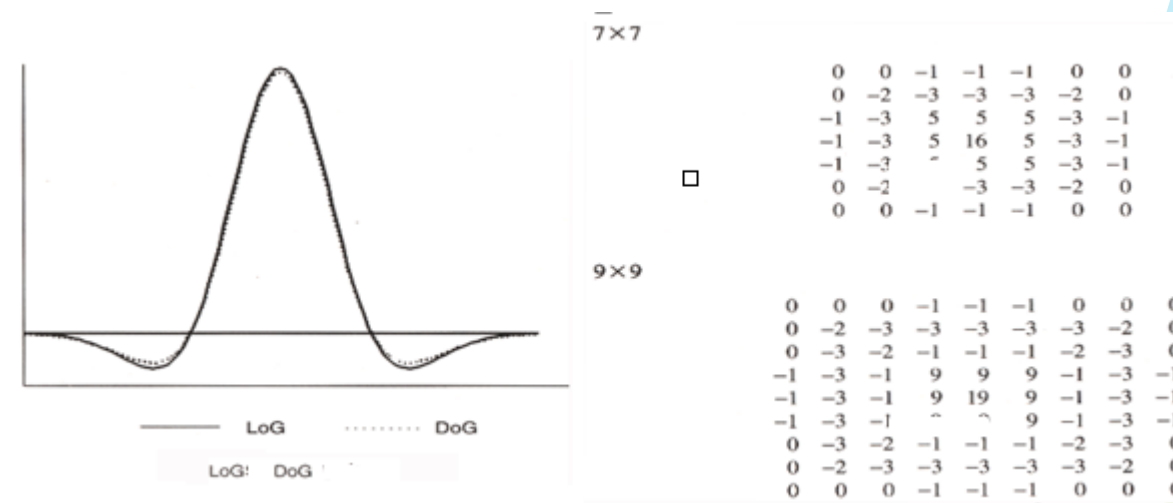


7X7, $\sigma=1.0$

DoG(Difference of Gaussians)

- ▶ DoG(Difference of Gaussians)
 - ▶ Approximation of LoG (Cf. Heavy computation requirements of the LoG)
 - ▶ The ratio $\sigma_1 / \sigma_2 = 1.6$ results in a good approximation of LoG
 - ▶ LoG ($\sigma = 12.35$) vs. DoG ($\sigma_1 = 10 \quad \sigma_2 = 16$)
 - ▶ Can specify the width of edges to detect by varying σ_1 and σ_2 .
 - ▶ The 9x9 mask will detect wider edges than 7x7 mask.

$$DoG(x, y) = \frac{e^{-\frac{(x^2+y^2)}{2\sigma_1^2}}}{2\pi\sigma_1^2} - \frac{e^{-\frac{(x^2+y^2)}{2\sigma_2^2}}}{2\pi\sigma_2^2}$$



DoG(Difference of Gaussians)

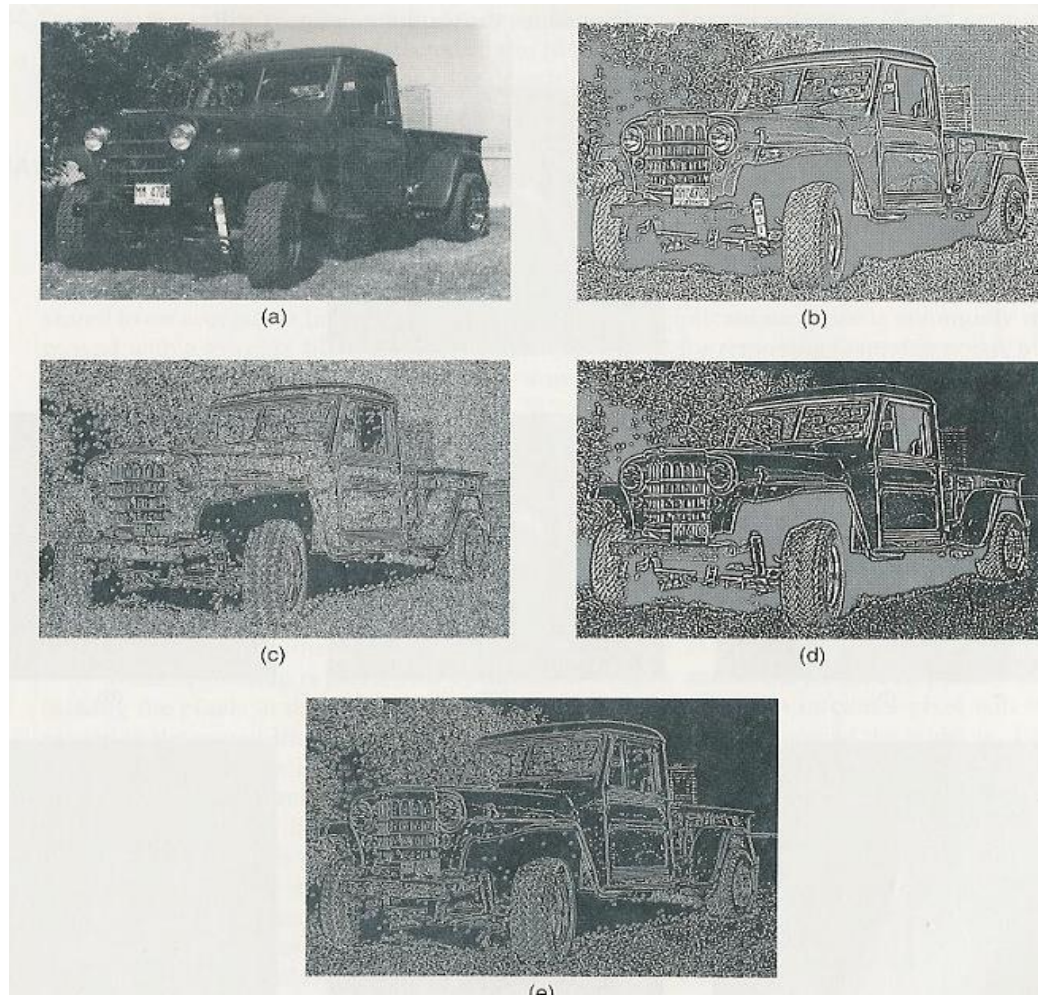


Fig. 3.23 (a) Original, (b) 7x7 Dog, (c) zero-crossing, (d) 9x9 Dog, and (e) zero-crossing.

Color Edge Detection

- ▶ Color Edge Detection

- ▶ Depends on your definition of an edge.

- ▶ (Def.1) the discontinuity in an image's luminance => Edge detection would be done on the intensity channel of color image in HSU space.
 - ▶ (Def. 2) the discontinuity in RGB channels => Edge detection is performed on each color components.
 - ▶ (Def. 3) Performed each component; the components are summed to create the gray scale map.

$$G(x, y) = \sqrt{G_{red}^2 + G_{green}^2 + G_{blue}^2}$$

- ▶ The best color edge detection => Depends on the application.

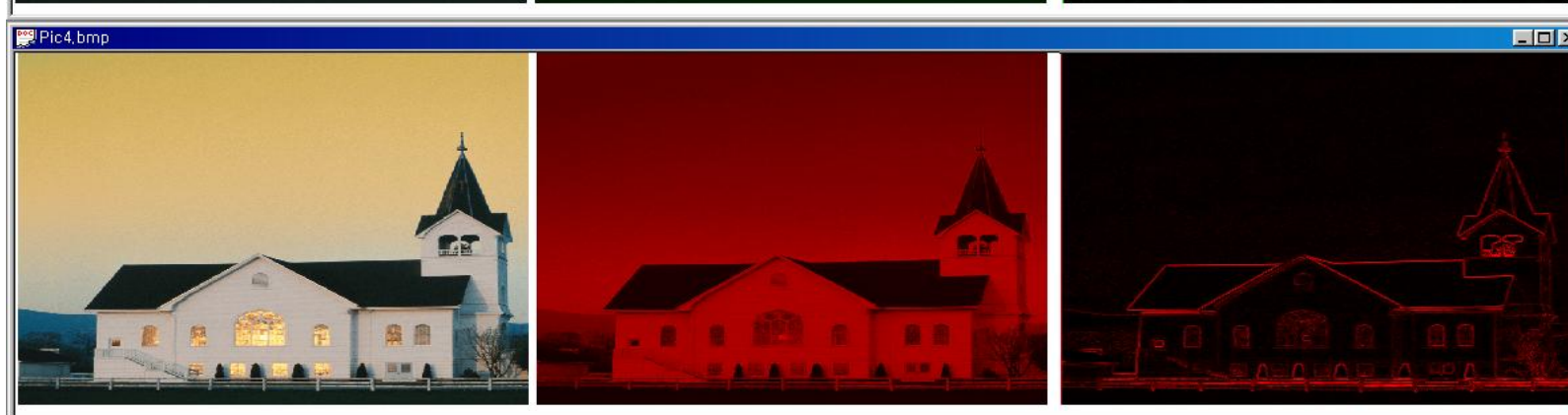
Color Edge Detection



(a) Blue Channel



(b) Green Channel



(c) Red Channel

Median Filtering

- ▶ Median filtering

- ▶ An effective method for removing impulse noise
- ▶ Preserve sharp edges(high frequency information) and detail
- ▶ Lowpass filter
 - ▶ Suited for removing Gaussian noise, but not impulse noise
- ▶ Median filter is superior to lowpass filter when dealing with edges

Median Filtering

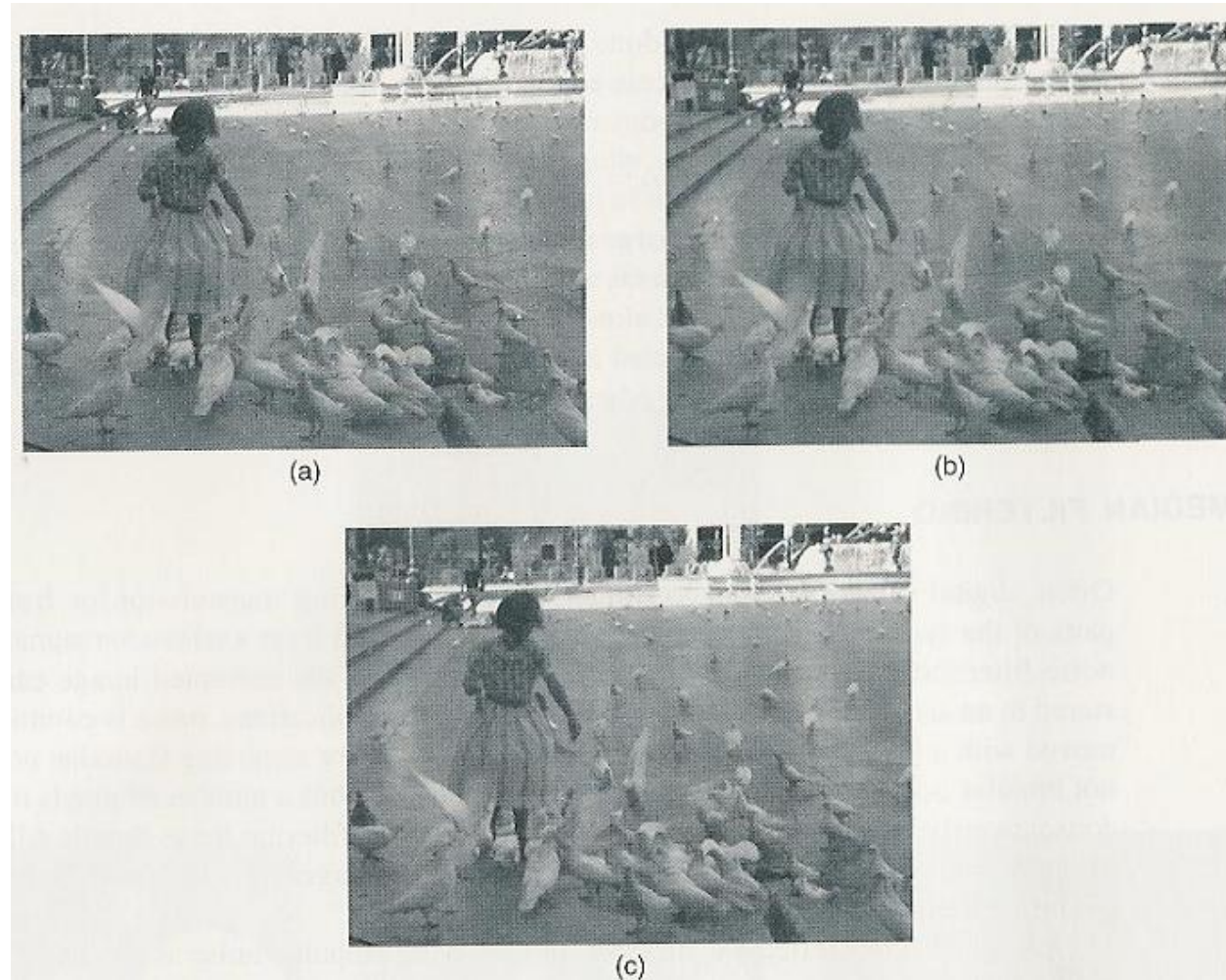
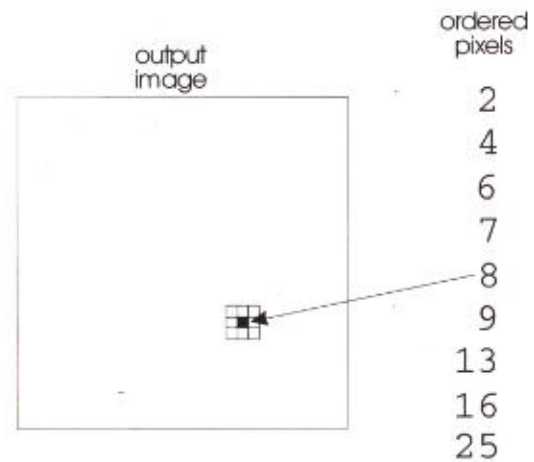
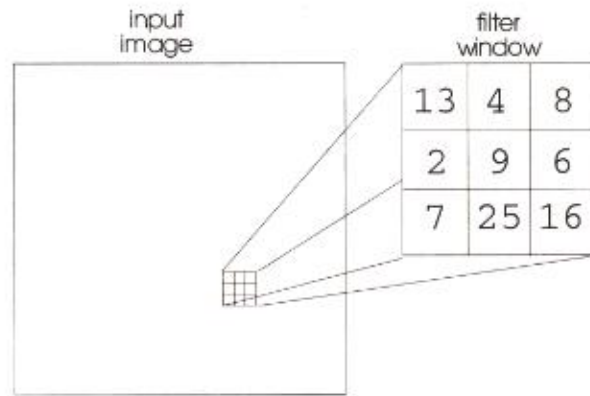
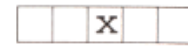


Fig. 3.25 (a) Noisy image, (b) low-pass filtered, and (c) median filtered.

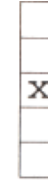
Median Filtering



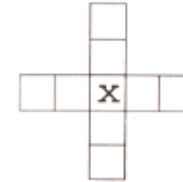
Median Filtering Method



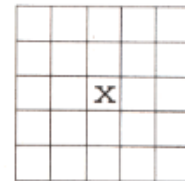
horizontal



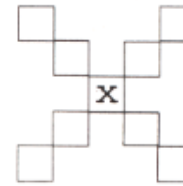
vertical



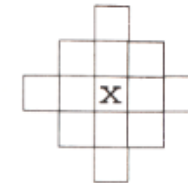
cross



block



x



diamond

Median Filtering



**Original image
(Noise Added)**



Median Filter(3x3)



Median Filter(5x5)



Median Filter(9x9)

Median Filter



(a) Original image

(b) Median filtered with a 3 X 3 mask

(c) Median filtered with a 5 X 5 mask

(d) Median filtered with a 7 X 7 mask

Color Median Filtering

- ▶ Color Median Filtering with separate R,G,B
 - ▶ Apply median filter to each R, G, B, separately.
 - ▶ (Problem) Lose any correlation between color components.
 - ▶ Ex. Horizontal median filter of width 5 => Can not remove unwanted yellow pixel.

						edge				
Red	{	0	0	255	0	0	255	255	255	}
Green	{	255	255	255	255	255	0	0	0	}
Blue	{	0	0	0	0	0	0	0	0	}
Composite Color	{	Green	Yellow	Green			Red			}

						edge					
Red	{	0	0	0	0	255	255	255	255	}	
Green	{	255	255	255	255	255	0	0	0	}	
Blue	{	0	0	0	0	0	0	0	0	}	
Composite Color	{	Green				Yellow	Red				}

Color Median Filtering

- ▶ Color median filtering using the unique property.

- ▶ A unique property of median values when filtering color image

$$\sum_{i=1}^N |x_{med} - x_i| \leq \sum_{i=1}^N |y - x_i|$$

- ▶ where N is the number of elements e.g., 9 for a 3 x 3 median filter; y is an arbitrary value in the window.

- ▶ (Idea) Break x, y in the unique property into red, green, blue components as follow:

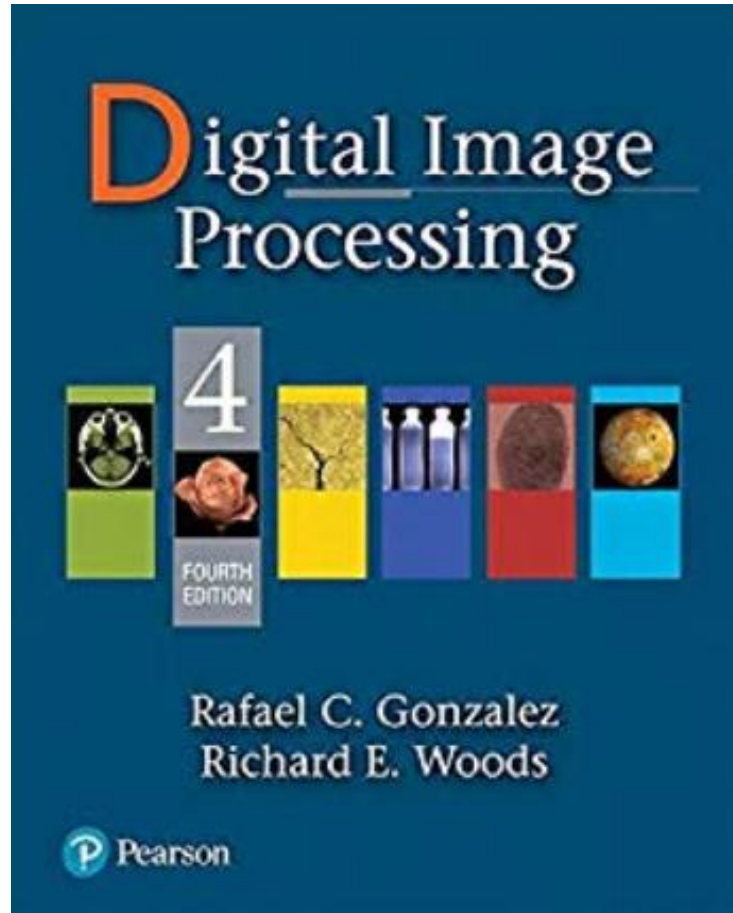
$$Dist_i = \sum_{i=1}^N |red_i - red_j| + \sum_{i=1}^N |green_i - green_j| + \sum_{i=1}^N |blue_i - blue_j|$$

- ▶ The pixel x_i with smallest $Distance_i$ is the filter output, where $x_i = \{red_i, green_i, blue_i\}$
 - ▶ (Algorithm): Computation of the color median output
 - ▶ Start with N elements in the filter window
 - ▶ Compute $Distance_i$, and find minimum $Distance_i$
 - ▶ Return x_i with minimum $Distance_i$

Minimum/Maximum Filtering

- ▶ Minimum/maximum filtering
 - ▶ Very similar to median filtering.
 - ▶ Max/min window value is used. Cf. Central pixel in median window value.
 - ▶ Effective at removing extreme impulse noise.
 - ▶ Min filter: removes white spikes; overall intensity will be reduced
 - ▶ Max: removes dark spikes; overall intensity is increased
 - ▶ Both the min and max filters fail in removing mixed impulse noises.
- ▶ Cascade of max min filters: can remove “salt and pepper noise”
 - ▶ Closing filter: min $\rightarrow$ max
 - ▶ Opening filter: max $\rightarrow$ min

Reference



Question



Thank you

